Level: Bachelor Semester: Spring Year : 2024
Programme: BE Full Marks: 100
Course: Problem Solving Techniques (New) Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain why there are infinitely many prime numbers. 7

   b) Assume that k is a positive integer. If $(k+1)$ letters are delivered to k mailboxes, then show that one mailbox contain at least one letters. 8

2. a) A right circular cone has a cube inscribed in it. If the radius of the cone is 1, and its height is 3, then what is the volume of the cube? 8

   OR

   Of all parallelograms with a given perimeter, which has the greatest area?

   b) Let l and m be two lines in the plane that are skew to each other (i.e. they intersect at a single point X). Let P be a point (other than X) on the line l. Using a ruler and compass, construct a circle that is tangent to both lines and passes through P. 7

3. a) Suppose that you have 37 envelopes and you address 37 letters to go with them. Closing your eyes, you randomly stuff one letter into each envelope. What is the probability that precisely two letters are in the wrong envelopes and all others in the correct envelope? 7

   b) Solve the following crypto arithmetic problem: 8

$$\begin{array}{r} SEVEN \\ +EIGHT \\ \hline TWELVE \end{array}$$

4. a) Suppose that you have 9 pearls. They all look the same, but 8 of have equal weight and one is different. The odd pearl is either heavier or lighter; you do not know which. The only equipment that you have at hand is a balance scale. How can you use the scale to find the odd pearl in just three weighing? 8

   OR

   Develop the strategy and write an algorithm for an 8×8 magic square and construct it.